

Integration Methods and ODE Solvers

❑ **Objective:** To find a **numerical solution** for the state-space equation below:

$$\left\{ \begin{array}{l} \frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \in \mathcal{R}^m \end{array} \right.$$

[t₀, t_f]

✓ often called **Initial Value Problem (IVP)**, i.e., given this starting point, does there exist a solution, and is it unique?

➤ We want to obtain numerical solution for $\mathbf{x}(t)$ on time interval $t \in [t_0, t_f]$

❖ Recall from Calculus:

❑ **Picard–Lindelöf Theorem:**

✓ if the derivative function $\mathbf{f}$ is continuous in t and uniform Lipschitz on $\mathbf{x}$, there exists a unique solution to $\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t)$ for any $\mathbf{x}_0 \in \mathcal{R}^m$



✓ crucial for ensuring **existence and uniqueness** of the solutions and know **convergence** of approximation methods

Integration Methods and ODE Solvers

□ **Objective:** To find a **numerical solution** for the state-space equation:

❖ Recall from Calculus:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) \quad \mathbf{x}(t_0) = \mathbf{x}_0 \in \mathcal{R}^m$$

□ Uniform Lipschitz Condition:

✓ if there exists a **Lipschitz constant** L such that

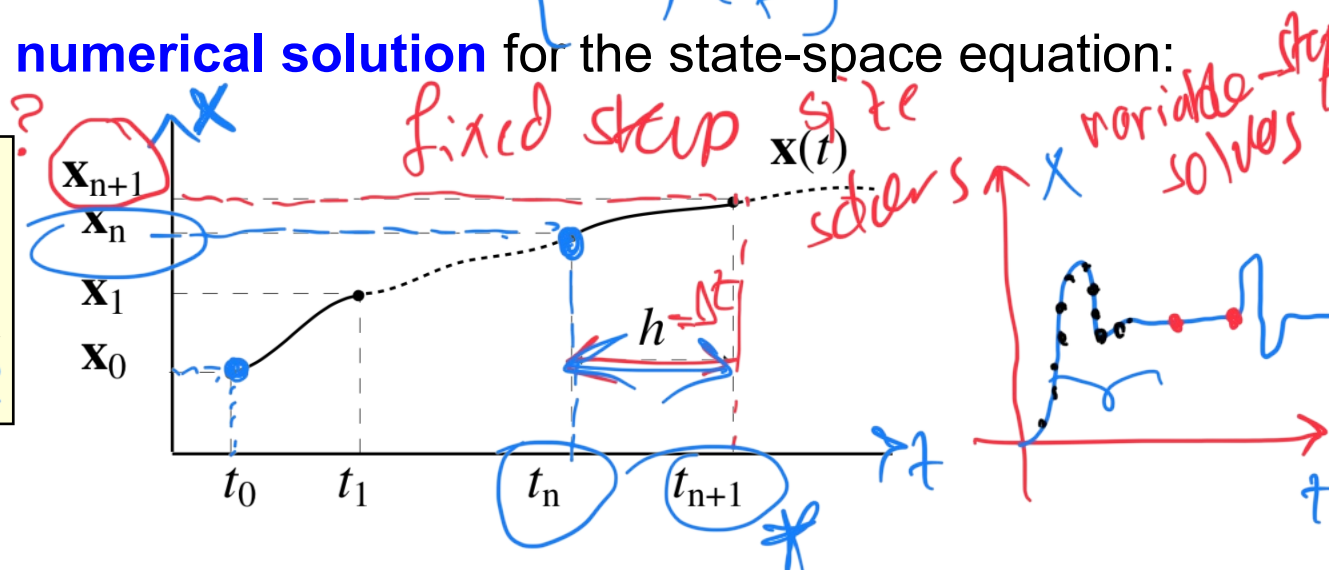
$$\|\mathbf{f}(\mathbf{x}_1, t) - \mathbf{f}(\mathbf{x}_2, t)\| \leq L \cdot \|\mathbf{x}_1 - \mathbf{x}_2\| \quad \text{for all } \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{R}^m$$

❖ **What are the implications?**

Integration Methods and ODE Solvers

❑ **Objective:** To find a **numerical solution** for the state-space equation:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \in \mathcal{R}^m \end{cases}$$



❖ **Define:**

- ✓ $\mathbf{x}_n$ is the solution at point t_n (i.e., current step)
- * ✓ $\mathbf{x}_{n+1}$ is the numerical solution/approximation at point t_{n+1} (i.e., next step)
- ✓ h is the time step between the points t_n and t_{n+1} (i.e., Δt)

Note: numerical solution is available only at certain grid points!

$$\int_{x_n}^{x_{n+1}} dx = x_{n+1} - x_n = \int_{t_n}^{t_{n+1}} f(x, t) dt$$

General form of solution

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \times \Phi(\mathbf{x}, t, h)$$

Φ is called the method or update formula

Handwritten notes include: $x_{n+1} = x_n + \int_{t_n}^{t_{n+1}} f(x, t) dt$ and $x_{n+1} = x_n + h \times \Phi(\mathbf{x}, t, h)$.

Truncation Error

round off error 3-12667

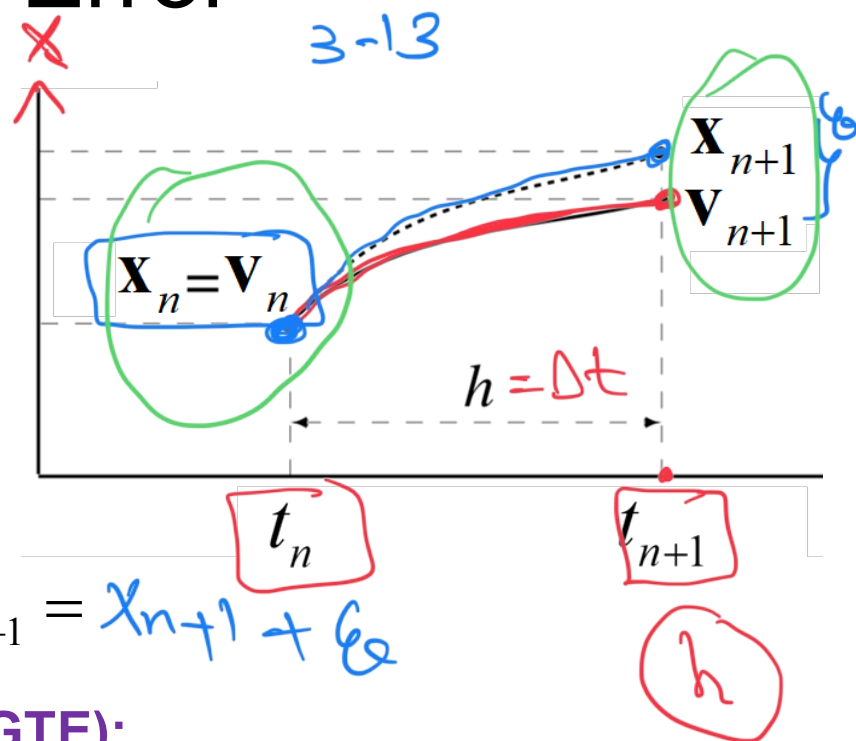
❑ **Remember:** Objective was to compute $\mathbf{x}_{n+1}$ accurately!

➤ Assume $\mathbf{v}(t)$ is the “valid/true” solution that goes through $\mathbf{x}_n$

➔ $\mathbf{x}_n = \mathbf{v}_n = \mathbf{v}(t_n)$

➤ What about $\mathbf{v}(t_{n+1})$?

➔ $\mathbf{v}_{n+1} = \mathbf{x}_{n+1} + \epsilon$



❖ Local/Global Truncation Error (LTE/GTE):

✓ LTE is defined for one step as:

$$\text{LTE} = \|\mathbf{x}_{n+1} - \mathbf{v}_{n+1}\|$$

✓ LTE per-step:

$$\tau = \frac{\|\mathbf{x}_{n+1} - \mathbf{v}_{n+1}\|}{h}$$

✓ GTE:

LET accumulated over $1/h$ steps!

❖ Order of the Method:

* Method $\Phi(\mathbf{x}, t, h)$ has order $p > 0$ if there exists a positive bounded constant C such that

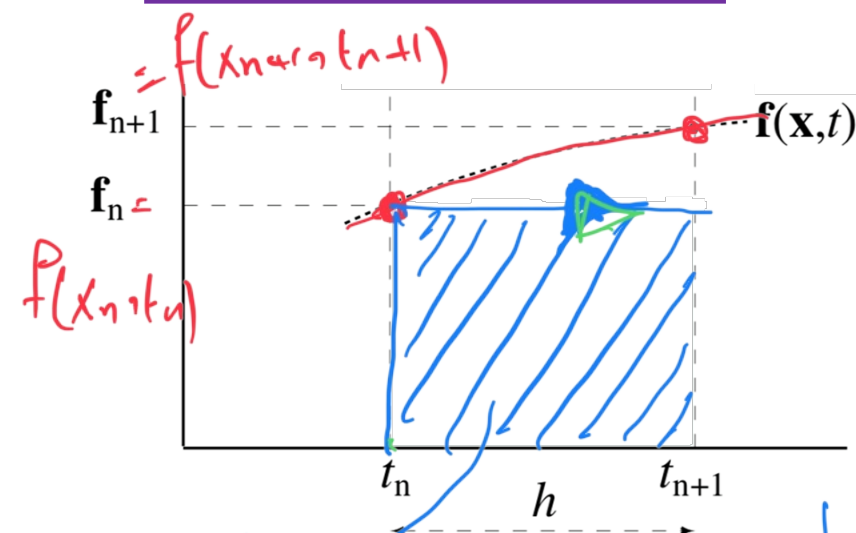
$$\tau \leq C \cdot h^p$$

$O(h^p)$
 $p=2$

Integration Methods and ODE Solvers

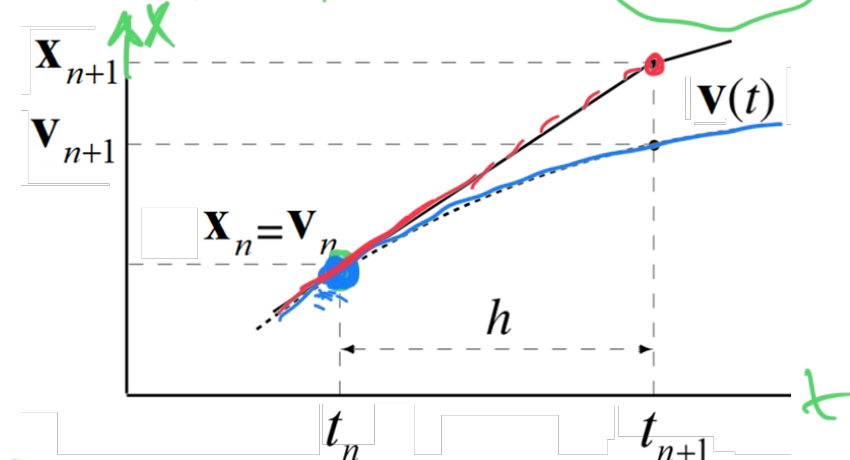
❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$.

❖ Forward Euler Method:



$$x_{n+1} = x_n + h \cdot \phi$$

$$\left(\frac{\int f dt}{h} \right)$$



$$A = \int f dt = h \times f_n$$

$$\frac{dx}{dt} = f(x_n, t_n)$$

$$x_{n+1} = x_n + h f(x_n, t_n)$$

$$\Phi(\mathbf{x}, t, h) = f(\mathbf{x}_n, t_n)$$

$$x_{n+1} = x_n + h \times f(x_n, t_n)$$

➤ Is the method explicit or implicit?

Explicit ✓

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

❖ Order of Forward Euler Method:

$$\frac{dx}{dt} = f(x, t)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \times \mathbf{f}(\mathbf{x}_n, t_n)$$

➤ Assume $\mathbf{v}_n = \mathbf{x}_n$ and $\frac{d\mathbf{v}(t_n)}{dt} = \mathbf{v}'_n = \mathbf{x}'_n$

✓ Use Taylor's series:

~~LTF~~
per-step: $\tau = \frac{\|\mathbf{x}_{n+1} - \mathbf{v}_{n+1}\|}{h}$

$$\tau \leq C \cdot h^p$$

$$\mathbf{v}_{n+1} = \mathbf{v}(t_n + h) = \mathbf{v}(t_n) + h \frac{d\mathbf{v}(t_n)}{dt} + \frac{h^2}{2!} \frac{d^2\mathbf{v}(t_n)}{dt^2} + \dots$$

$t_n < \xi < t_{n+h}$

$$\mathbf{v}_{n+1} = \mathbf{x}_n + h \times \mathbf{f}_n + \frac{h^2}{2!} \frac{d^2\mathbf{v}(\xi)}{dt^2}$$

$$\frac{|\mathbf{v}_{n+1} - \mathbf{x}_{n+1}|}{h} = \frac{h}{2} \frac{d^2\mathbf{v}(\xi)}{dt^2} \Rightarrow \tau = \frac{h}{2} \left| \frac{d^2\mathbf{v}(\xi)}{dt^2} \right| \leq C \cdot h$$

➤ What does being first-order ($O(h^1)$) mean?

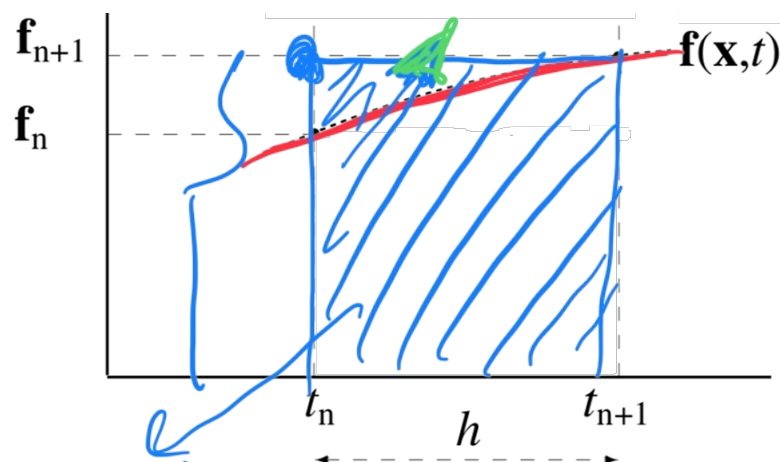
$$p=1$$

$$\tau \leq C \cdot h^1 \leq C$$

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

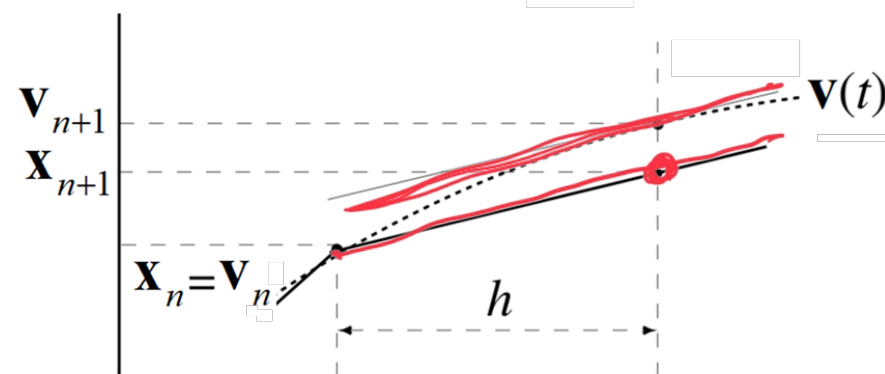
❖ Backward Euler Method:



$$A = \int f dt = h \times f_{n+1}$$

$$x_{n+1} = x_n + \int f dt$$

$$x_{n+1} = x_n + h \times \Phi$$



$$x_{n+1} - x_n - h f(x_{n+1}, t_{n+1}) = F(x_{n+1}) = 0$$

$$x_{n+1} = x_n + h \times f(x_{n+1}, t_{n+1})$$

$$\Phi(\mathbf{x}, t, h) = f(x_{n+1}, t_{n+1})$$

➤ Is the method explicit or implicit? *implicit*

➤ What is the order of BE method? $O(h^1)$

unknown
 x_{n+1} is on both
of the equation

Integration Methods and ODE Solvers

❖ Newton-Raphson Method:

❑ Objective is to find (iteratively) the solution of a nonlinear equation expressed as

$$f(x) = c$$



➤ Suppose $x^{(0)}$ is an initial estimate of the solution, and $\Delta x^{(0)}$ is a small deviation from the correct solution; therefore:

$$f(x^{(0)} + \Delta x^{(0)}) = c$$

➤ The left-side expression can be expanded using Taylor's series:

$$\cancel{f(x^{(0)})} + \left(\frac{df}{dx}\right)^{(0)} \Delta x^{(0)} + \frac{1}{2!} \left(\frac{d^2 f}{dx^2}\right)^{(0)} (\Delta x^{(0)})^2 + \dots = c - \cancel{f(x^{(0)})}$$

➤ Assuming the error $\Delta x^{(0)}$ is very small, solve for where $\Delta c^{(0)} = c - f(x^{(0)})$

$$\Delta x^{(0)} \approx \left[\left(\frac{df}{dx} \right)^{(0)} \right]^{-1} \Delta c^{(0)}$$

➤ Update the value of new estimate as $x^{(1)} = x^{(0)} + \Delta x^{(0)}$

➤ Continue iterations until error becomes less than threshold $|x^{(k+1)} - x^{(k)}| \leq \epsilon$

Integration Methods and ODE Solvers

❖ Newton-Raphson Method:

❑ Objective is to find (iteratively) the solution of a nonlinear equation $f(x) = c$

➤ The successive procedure can be summarized as

$$\Delta c^{(k)} = c - f(x^{(k)}) \longrightarrow \Delta x^{(k)} = \left[\left(\frac{df}{dx} \right)^{(k)} \right]^{-1} \Delta c^{(k)} \longrightarrow x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \longrightarrow |x^{(k+1)} - x^{(k)}| \leq \varepsilon \longrightarrow \Delta c^{(k)}$$

➤ What about for a linear system? Is BE implicit or explicit?

$$\frac{dx}{dt} = AX$$

$$x_{n+1} = x_n + h \times A x_{n+1}$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \times \mathbf{f}(\mathbf{x}_{n+1}, t_{n+1})$$

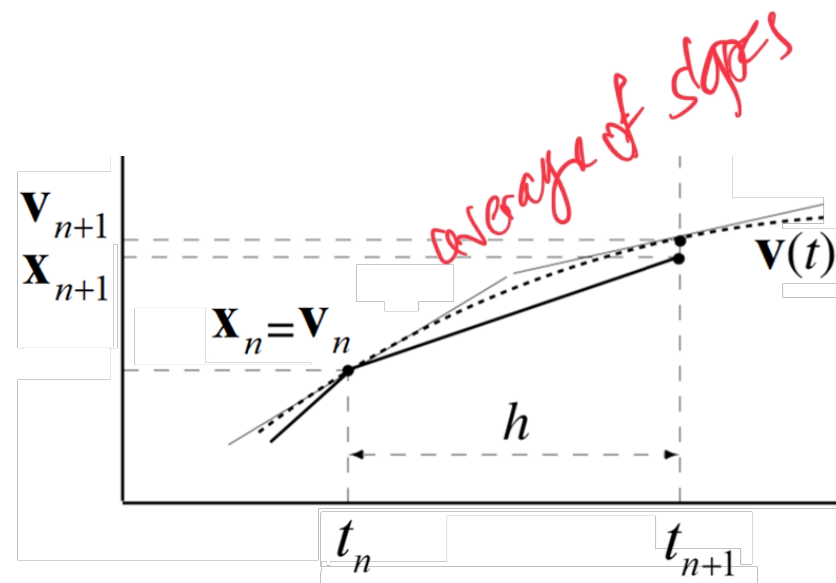
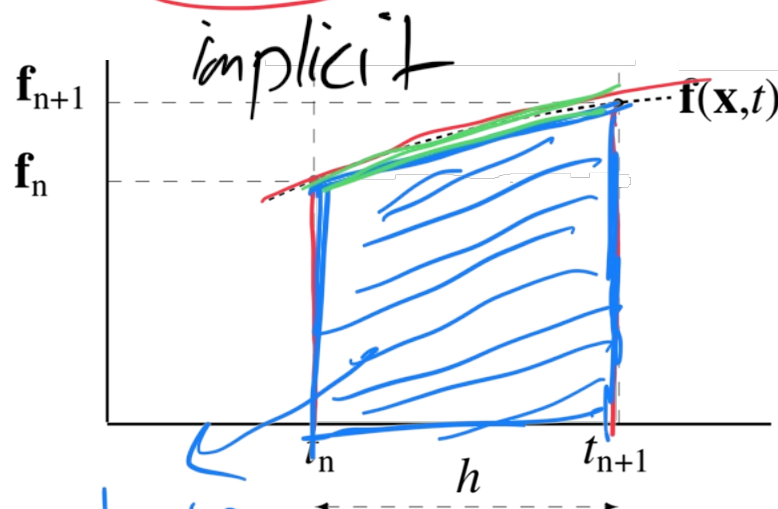
$$x_{n+1} - hA x_{n+1} = x_n \Rightarrow (I - hA) x_{n+1} = x_n \quad *$$

$$x_{n+1} = (I - hA)^{-1} x_n$$

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

❖ Trapezoidal Method:



$$A = \frac{h}{2} (f_n + f_{n+1})$$

$$X_{n+1} = X_n + \frac{h}{2} (f_n + f_{n+1})$$

$$\Phi(\mathbf{x}, t, h) = \frac{f(\mathbf{x}_n, t_n) + f(\mathbf{x}_{n+1}, t_{n+1})}{2}$$

$$X_{n+1} = X_n + \frac{h}{2} (f(\mathbf{x}_n, t_n) + f(\mathbf{x}_{n+1}, t_{n+1}))$$

➤ Is the method explicit or implicit?

implicit

➤ What is the order of Trapezoidal method?

$O(h^2)$

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

❖ Explicit Trapezoidal (Heun's) Method:

ODE 2

✓ avoids the implicit formulation using an **explicit predictor-corrector** strategy

1) **Predictor (Euler's step):**

$$\mathbf{x}_{n+1} \approx \mathbf{x}^* = \mathbf{x}_n + h \times \mathbf{f}_n$$

$\mathbf{x}_{n+1}$
 $\mathcal{I}(\mathbf{x}_n + h \mathbf{f}_n, t_{n+1})$

2) **Corrector (explicit trapezoidal update):**

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{2} \times (\mathbf{f}(\mathbf{x}_n, t_n) + \mathbf{f}(\mathbf{x}^*, t_{n+1}))$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \Phi$$

$$\alpha_1 + \alpha_2 = 1$$

$$\Phi(\mathbf{x}, t, h) = \alpha_1 \mathbf{k}_1 + \alpha_2 \mathbf{k}_2$$

$$\begin{cases} \alpha_1 = 1/2 \\ \alpha_2 = 1/2 \end{cases}$$

$$\begin{cases} \mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n) \\ \mathbf{k}_2 = \mathbf{f}(\mathbf{x}_n + h \mathbf{f}_n, t_n + h) \end{cases}$$

➤ Is the method explicit or implicit? *Explicit*

➤ What is the order of Heun's method? $\mathcal{O}(h^2)$

r -Stage Runge-Kutta Methods

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

General form of solution:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \times \Phi(\mathbf{x}, t, h)$$

❖ **General r -Stage Runge-Kutta:**

$$\Phi(\mathbf{x}, t, h) = \sum_{i=1}^r \alpha_i \mathbf{k}_i$$

$$\sum_{i=1}^r \alpha_i = 1$$

$$\alpha_1 \mathbf{k}_1 + \alpha_2 \mathbf{k}_2 + \dots + \alpha_r \mathbf{k}_r$$

➤ **Explicit Methods:**

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n)$$

$$\mathbf{k}_i = \mathbf{f}\left(\mathbf{x}_n + h \sum_{j=1}^{i-1} \lambda_{ij} \mathbf{k}_j, t_n + \mu_i h\right)$$

$$i \in \{2, 3, \dots, r\}$$

$$\mathbf{k}_3 = \mathbf{f}\left(\mathbf{x}_n + h \sum_{j=1}^2 \lambda_{3j} \mathbf{k}_j, t_n + \mu_3 h\right)$$

$$\lambda_{31} \mathbf{k}_1 + \lambda_{32} \mathbf{k}_2$$

➤ **Implicit Methods:**

$$\mathbf{k}_i = \mathbf{f}\left(\mathbf{x}_n + h \sum_{j=1}^r \lambda_{ij} \mathbf{k}_j, t_n + \mu_i h\right) \quad i \in \{1, 2, \dots, r\}$$

r - Stage Runge-Kutta Methods

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

➤ **Explicit Methods:**

$$\Phi(\mathbf{x}, t, h) = \sum_{i=1}^r \alpha_i \mathbf{k}_i \quad \sum_{i=1}^r \alpha_i = 1$$

$\alpha_1, \alpha_2, \dots, \alpha_r$

$$x_{n+1} = x_n + h \phi$$

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n) \quad i \in \{2, 3, \dots, r\}$$

$$\mathbf{k}_i = \mathbf{f}\left(\mathbf{x}_n + h \sum_{j=1}^{i-1} \lambda_{ij} \mathbf{k}_j, t_n + \mu_i h\right)$$

❑ **Classical RK Formulas:** $r=3, p=3$

$$\Phi = \frac{1}{6} \mathbf{k}_1 + \frac{4}{6} \mathbf{k}_2 + \frac{1}{6} \mathbf{k}_3$$

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n)$$

$$\mathbf{k}_2 = \mathbf{f}\left(\mathbf{x}_n + \frac{1}{2} h \mathbf{k}_1, t_n + \frac{1}{2} h\right)$$

$$\mathbf{k}_3 = \mathbf{f}\left(\mathbf{x}_n + h(2\mathbf{k}_2 - \mathbf{k}_1), t_n + h\right)$$

$$\begin{cases} \alpha_1 = 1/6 \\ \alpha_2 = 4/6 \\ \alpha_3 = 1/6 \end{cases} \quad \alpha_1 + \alpha_2 + \alpha_3 = 1$$

$$\begin{cases} \lambda_{21} = 1/2 \\ \lambda_{31} = -1 \\ \lambda_{32} = 2 \end{cases} \quad \begin{cases} \mu_2 = 1/2 \\ \mu_3 = 1 \end{cases}$$

r - Stage Runge-Kutta Methods

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

➤ **Explicit Methods:**

$$\Phi(\mathbf{x}, t, h) = \sum_{i=1}^r \alpha_i \mathbf{k}_i \quad \sum_{i=1}^r \alpha_i = 1$$

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n) \quad i \in \{2, 3, \dots, r\}$$

$$\mathbf{k}_i = \mathbf{f}\left(\mathbf{x}_n + h \sum_{j=1}^{i-1} \lambda_{ij} \mathbf{k}_j, t_n + \mu_i h\right)$$

❑ **MATLAB's ODE3 (Bogacki–Shampine) Formula:**

$$\Phi = \frac{2}{9} \mathbf{k}_1 + \frac{3}{9} \mathbf{k}_2 + \frac{4}{9} \mathbf{k}_3$$

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n)$$

$$\mathbf{k}_2 = \mathbf{f}\left(\mathbf{x}_n + \frac{1}{2} h \mathbf{k}_1, t_n + \frac{1}{2} h\right)$$

$$\mathbf{k}_3 = \mathbf{f}\left(\mathbf{x}_n + \frac{3}{4} h \mathbf{k}_2, t_n + \frac{3}{4} h\right)$$

$$\alpha_1 + \alpha_2 + \alpha_3 = 1$$

r - Stage Runge-Kutta Methods

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n) \quad \mathbf{k}_i = \mathbf{f}\left(\mathbf{x}_n + h \sum_{j=1}^{i-1} \lambda_{ij} \mathbf{k}_j, t_n + \mu_i h\right) \quad i \in \{2, 3, \dots, r\}$$

❑ **Classical RK Formulas:**

$r = 4, p = 4$

$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 1$

Runge's Coefficients (1895)

$$\Phi = \frac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$$

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n)$$

$$\mathbf{k}_2 = \mathbf{f}\left(\mathbf{x}_n + \frac{1}{2}h\mathbf{k}_1, t_n + \frac{1}{2}h\right)$$

$$\mathbf{k}_3 = \mathbf{f}\left(\mathbf{x}_n + \frac{1}{2}h\mathbf{k}_2, t_n + \frac{1}{2}h\right)$$

$$\mathbf{k}_4 = \mathbf{f}(\mathbf{x}_n + h\mathbf{k}_3, t_n + h)$$

Kutta's Coefficients (1901)

$$\Phi = \frac{1}{8}(\mathbf{k}_1 + 3\mathbf{k}_2 + 3\mathbf{k}_3 + \mathbf{k}_4)$$

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n, t_n)$$

$$\mathbf{k}_2 = \mathbf{f}\left(\mathbf{x}_n + \frac{1}{3}h\mathbf{k}_1, t_n + \frac{1}{3}h\right)$$

$$\mathbf{k}_3 = \mathbf{f}\left(\mathbf{x}_n + \frac{1}{3}h\mathbf{k}_2, t_n + \frac{1}{3}h\right)$$

$$\mathbf{k}_4 = \mathbf{f}(\mathbf{x}_n + h(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3), t_n + h)$$

r - Stage Runge-Kutta Methods

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

❖ **Maximum Order of r - Stage Runge-Kutta Methods (Butcher, 1960s):**

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(\mathbf{x}_n, t_n) & i \in \{2, 3, \dots, r\} \\ \mathbf{k}_i &= \mathbf{f}\left(\mathbf{x}_n + h \sum_{j=1}^{i-1} \lambda_{ij} \mathbf{k}_j, t_n + \mu_i h\right) \end{aligned}$$

$$\left\{ \begin{array}{l} 1 \leq r \leq 4, p^* = r \\ 5 \leq r \leq 7, p^* = r - 1 \\ 8 \leq r \leq 9, p^* = r - 2 \\ 10 \leq r, p^* = r - 3 \end{array} \right.$$

Handwritten notes:

- $\left\{ \begin{array}{l} r=1 \rightarrow p=1 \\ r=2 \rightarrow p=2 \\ r=3 \rightarrow p=3 \end{array} \right.$
- $\left\{ \begin{array}{l} r=4 \rightarrow p=4 \\ r=5 \rightarrow p=4 \\ r=6 \rightarrow p=5 \\ r=7 \rightarrow p=6 \end{array} \right.$

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

❖ Linearly Implicit Methods (Rosenbrock's Methods):

➤ Recall Implicit Euler:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \mathbf{f}(\mathbf{x}_{n+1}, t_{n+1}) = \mathbf{x}_n + h \mathbf{f}_{n+1}$$

$$\frac{dx}{dt} = f(x, t)$$

✓ If $\mathbf{f}(\mathbf{x}, t)$ was linear we could solve directly for $\mathbf{x}_{n+1}$

- For **stiff problems**, explicit RK methods require prohibitively small step sizes
- Fully **implicit RK methods** are expensive; must solve nonlinear equations
- **Rosenbrock methods** are a compromise: replace the **nonlinear** solves with **linear solves** using the Jacobian

➤ In general case, we can approximate $\mathbf{f}_{n+1}$ as:

$$\mathbf{f}(\mathbf{x}_{n+1}, t_{n+1}) \approx \mathbf{f}(\mathbf{x}_n, t_n) + \frac{\partial \mathbf{f}_n}{\partial \mathbf{x}} (\mathbf{x}_{n+1} - \mathbf{x}_n) + \frac{\partial \mathbf{f}_n}{\partial t} (t_{n+1} - t_n)$$

$$\mathbf{f}_{n+1} \approx \mathbf{f}_n + \mathbf{J}_x (\mathbf{x}_{n+1} - \mathbf{x}_n) + \mathbf{J}_t h$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h [\mathbf{f}_n + \mathbf{J}_x \mathbf{x}_{n+1} - \mathbf{J}_x \mathbf{x}_n + \mathbf{J}_t h]$$

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

❖ Linearly Implicit Methods (Rosenbrock's Methods):

ode 23s

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h[\mathbf{f}_n + \mathbf{J}_x \mathbf{x}_{n+1} - \mathbf{J}_x \mathbf{x}_n + \mathbf{J}_t h]$$

group terms

$$\begin{aligned} x_{n+1} &= x_n + h f_n + h J_x x_{n+1} - h J_x x_n + h J_t h \\ (I - h J_x) x_{n+1} &= (I - h J_x) x_n + h (f_n + J_t h) \end{aligned}$$

➤ The general form of the methods of this type is:

$$x_{n+1} = x_n + (I - h J_x)^{-1} h (f_n + J_t h)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \Phi(\mathbf{x}, h, t), \quad \Phi(\mathbf{x}, h, t) = \sum_{i=1}^r \alpha_i \mathbf{k}_i$$

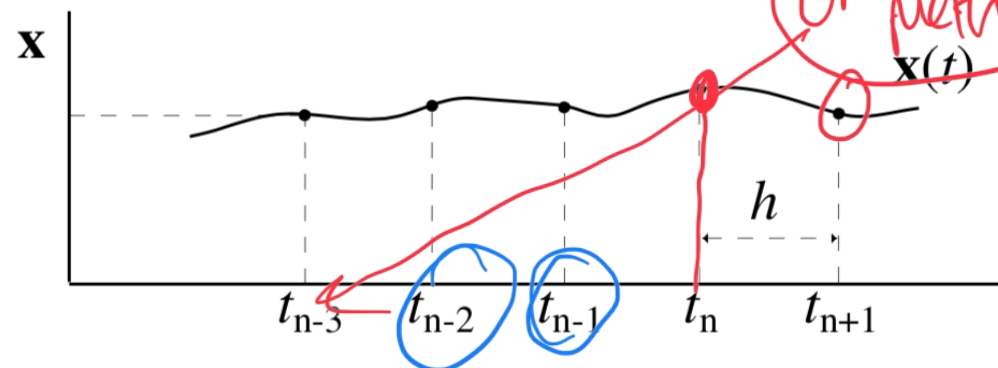
$$\mathbf{k}_i = (\mathbf{I} - h \mathbf{J}_x)^{-1} \left[\mathbf{f} \left(\mathbf{x}_n + h \sum_{j=1}^r \lambda_{ij} \mathbf{k}_j, t_n + \mu_i h \right), h d_i \mathbf{J}_t \right]$$

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

❖ Multi-Step Methods:

- ✓ Use several previous data for predicting $\mathbf{x}_{n+1}$



$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \in \mathbb{R}^m \end{cases}$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \mathbf{f}_n$$

Memory!

➤ General form of solution:

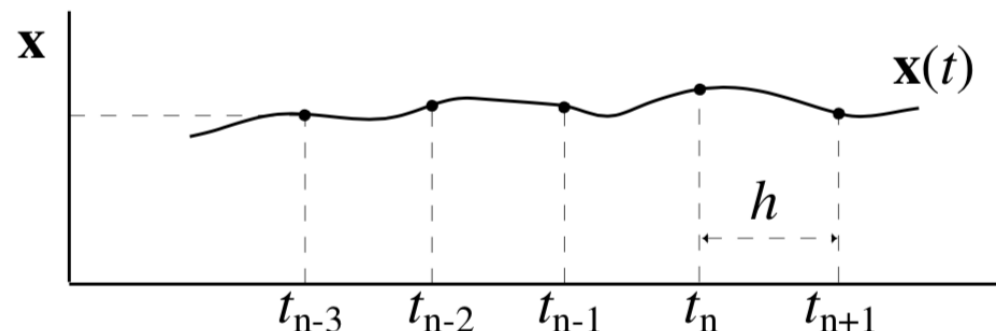
$$\mathbf{x}_{n+1} - \alpha_0 \mathbf{x}_n - \alpha_1 \mathbf{x}_{n-1} - \alpha_2 \mathbf{x}_{n-2} - \dots = h(\beta_{-1} \mathbf{f}_{n+1} + \beta_0 \mathbf{f}_n + \beta_1 \mathbf{f}_{n-1} + \beta_2 \mathbf{f}_{n-2} + \dots)$$

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

❖ k - Step Method:

✓ Use several previous data for predicting $\mathbf{x}_{n+1}$



➤ General form of solution:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \in \mathfrak{R}^m \end{cases}$$

$$\mathbf{x}_{n+1} = \sum_{i=0}^{k-1} \alpha_i \mathbf{x}_{n-i} + h \sum_{i=-1}^{k-1} \beta_i \mathbf{f}_{n-i}$$

$L_{n+1} = \mathcal{I}_h(\mathbf{x}_{n+1}, t_{n+1})$

➤ α_i and β_i depend on the method



✓ derived based on polynomial interpolation

{ If $\beta_{-1} = 0$ the method is **explicit** ✓
 If $\beta_{-1} \neq 0$ the method is **implicit** ✓



✓ polynomial order

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

❖ Examples of k -Step Methods:

Adams-Bashforth:
(k -step **Explicit**)

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \sum_{i=0}^{k-1} \beta_i \mathbf{f}_{n-i}$$

$$x_{n+1} = x_n + h\beta_0 f_n + h\beta_1 f_{n-1} + \dots$$

- ✓ one function evaluation per step
- small region of stability

✓ β_i are derived based on integration of the interpolating polynomial

Adams-Moulton:
(k -step **Implicit**)

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \sum_{i=-1}^{k-1} \beta_i \mathbf{f}_{n-i}$$

$$x_{n+1} = x_n + h\beta_{-1} f_{n+1} + h\beta_0 f_n + h\beta_1 f_{n-1} + \dots$$

- ✓ one more point than Adams-Bashforth (more accurate)
- ✓ large region of stability



$$\begin{cases} k = 1 \\ \beta_0 = 0 \end{cases}$$

$$\begin{cases} k = 1 \\ \beta_{-1} = \beta_0 = 1/2 \end{cases}$$

$\int(x_{n+1}, t_{n+1})$

Integration Methods and ODE Solvers

❑ **Objective:** To find the integral of $\mathbf{f}(\mathbf{x}, t)$ on interval $t \in [t_n, t_{n+1}]$:

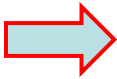
❖ Examples of Multi-Step Methods:

❖ **Backward Differentiation Formula (BDF) (Gear's method):**

$$\mathbf{x}_{n+1} = \sum_{i=0}^{k-1} \alpha_i \mathbf{x}_{n-i} + h \beta_0 \mathbf{f}_{n+1}$$

✓ α_i and β_0 are derived based on differentiation of the interpolating polynomial

- implicit, often used for stiff problems

❖ 1-step BDF: assume $\begin{cases} k = 1 \\ \beta_0 = 1 \end{cases}$ 

❖ 2-step BDF:

$$\mathbf{x}_{n+1} = \frac{4}{3} \mathbf{x}_n - \frac{1}{3} \mathbf{x}_{n-1} + h \frac{2}{3} \mathbf{f}(\mathbf{x}_{n+1}, t_{n+1})$$